

SHATHURYA PARAMANATHAN

Software Engineer

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PROFESSIONAL SUMMARY

Results-driven Software Engineer specialising in end-to-end delivery of scalable web platforms for Telecom, MedTech and HR-Tech domains. Proven ability to own the full software lifecycle — translating business requirements into robust architecture, building performant frontend and backend systems, and deploying secure, production-ready solutions to the cloud. Brings a strong engineering mindset to complex problems, with expertise in API design, data-driven dashboards, automation pipelines and AI-integrated applications. Committed to clean, maintainable code and continuous improvement within Agile, cross-functional teams.

EDUCATION

- **B.Sc.(Hons) in Information Technology** 2022 - 2026
University of Moratuwa, Colombo, Sri Lanka
L2S2 SGPA - **3.83 (Dean's list)**

PROFESSIONAL EXPERIENCE

- **COGNTIX — Software Engineer Intern** Feb 2025 – Jul 2025
Nallur, Jaffna, Sri Lanka

Software Engineering & Architecture

- Delivered production-grade full-stack applications in **React.js**, **Next.js**, **Node.js**, **Express.js**, and **Python** that now serve live customers across Telecom, MedTech, and HRTech domains.
- Translated business requirements into modular, scalable architecture, enabling new features to be added without rework and reducing long-term maintenance effort.
- Took ownership of the full **SDLC** - from requirements and architecture through development, testing, deployment, and production support - so features shipped end-to-end without hand-offs.
- Brought three products from concept to production - an AI-driven recruitment platform (Unblit), a Telecom administration portal (BeReload), and a multi-role MedTech platform (Avana) - giving the business three revenue-generating systems.
- Secured every application with **JWT**-based authentication and role-based access control (**RBAC**), ensuring sensitive data was only accessible to authorised roles.

Database & Data Management

- Designed and maintained **MongoDB** collections and document schemas, implementing efficient querying and indexing strategies to support scalable application performance.
- Introduced validation, error handling, and structured logging across backend services, reducing debugging effort and preventing invalid data from reaching production systems.

Cloud & Deployment

- Deployed and hardened production workloads on **AWS Lightsail** with server, networking, and **SSL/TLS** configuration, delivering secure, always-available environments for end users.
- Established repeatable deployment workflows and environment-based configuration, making releases predictable and reducing deployment risk.

Backend & API Engineering

- Built **RESTful** services in **Node.js** and **Express.js** that power multi-role workflows and dynamic data processing across all three products.
- Streamlined frontend–backend communication with **Axios**, improving data retrieval and state management so screens loaded with fewer, faster requests.
- Extended the backend with **Python** services for document processing and AI workflows, unlocking automation the products could not previously offer.
- Validated and documented every endpoint in **Postman**, giving frontend and backend teams a single source of truth and reducing integration bugs.
- Integrated third-party APIs with robust auth, rate-limit, and failure handling, keeping external dependencies from disrupting user experience.

Frontend Engineering

- Built administrative portals in **React.js** and **Next.js** serving 4 distinct user roles from a single codebase, reducing duplication and maintenance.
- Delivered real-time dashboards with advanced filtering, pagination, search, and **Excel** export, giving operations teams instant visibility and removing manual reporting.
- Created reusable, responsive components with **Material UI** and **Tailwind CSS** plus client-side validation, ensuring consistent UX across devices and faster feature delivery.
- Optimised rendering, state management, and API round-trips, delivering noticeably faster page loads on data-heavy screens.

AI & Automation Engineering

- Built AI-assisted document analysis workflows in **Python** using **PyMuPDF**, **pdfPlumber**, **Tesseract OCR**, and the **GPT-4 API**.
- Implemented regular-expression-based data masking to protect sensitive personal information before AI processing.
- Designed prompts and structured-output handling to extract insights and generate consistent evaluation results from document content.

Quality & Engineering Practices

- Performed testing, debugging, and issue resolution across application components to improve stability and reliability.
- Used **Git** and **GitHub** for version control, maintaining branches and pull requests to manage code changes and collaborate effectively.

- Collaborated with team members during development and maintained clean, organized, and maintainable code throughout the development process.

KEY PROJECTS

- 1. PolicyLens AI — AI-Powered Website Policy Analysis Platform (Ongoing)** GitHub
 - Developing an AI-powered platform using **Python**, **FastAPI**, **Playwright**, and **MongoDB** to crawl company websites, discover policy pages, and extract legal and compliance-related content.
 - Building an automated document-processing pipeline to identify **Privacy Policy**, **Terms & Conditions**, **Cookie Policy**, **Refund Policy**, and **Security Policy** pages and convert web content into structured documents.
 - Designing the system to analyse extracted policies and present potential privacy, security, and compliance concerns in clear language for **non-technical users**, with an emphasis on explainability and reliable AI-assisted decision support.
- 2. THRIFTING.LK — E-Commerce Platform for Pre-Owned Items** GitHub
 - Built a full-stack marketplace in **React**, **Express.js**, **Node.js**, and **MongoDB** that lets users list, browse, and buy pre-owned items, with **Firestore Authentication** protecting every buyer and seller session.
 - Designed **RESTful APIs** and **MongoDB** models for onboarding, listings, and order tracking, enabling smooth browsing across 8 product categories.
 - Delivered a **React** seller dashboard for inventory, orders, and profile management, reducing the effort needed to list and manage items end-to-end.
- 3. SL Flood Risk Predictor — Machine Learning Application** Live Demo | GitHub
 - Trained an **XGBoost** model in **Python** that predicts flood risk across Sri Lanka with $R^2 = 0.989$ and low MAE/RMSE, giving reliable early-warning signals from environmental, geographic, and infrastructure data.
 - Built a full preprocessing pipeline in **Pandas** and **Scikit-learn** and applied **SHAP** interpretability, revealing rainfall, elevation, and river distance as the dominant risk drivers so results are explainable to non-technical users.
 - Published the model as a live **Streamlit** dashboard, making real-time risk prediction and visualisation accessible to anyone with a browser.
- 4. Expense Tracker — Personal Finance Web Application** GitHub
 - Built a secure full-stack app on **Next.js** with **Clerk** authentication and a **Drizzle ORM** relational schema, giving users a private, reliable record of their transactions and categories.
 - Delivered a categorised transaction log and real-time analytics dashboard that surfaces spending patterns by category, helping users spot overspending and make better financial decisions.
 - Designed a clean, fully responsive interface with **Tailwind CSS**, providing a consistent experience on desktop and mobile.
- 5. Family Budget Planner — Full-Stack Personal Finance Management System (Ongoing)** GitHub
 - Developing a full-stack family finance platform using **ASP.NET Core**, **React**, **TypeScript**, **SQL Server**, and **Entity Framework Core** for managing budgets, expenses, savings, tasks, and bills.
 - Building secure **RESTful APIs** with **JWT authentication** and role-based authorization to support multiple family members with controlled access.
 - Implementing financial dashboards, budget tracking, and category-based analytics while following clean architecture and modern software engineering practices.

RESEARCH EXPERIENCE

- 1. AI-Powered Big Onion Yield Prediction — Final Year Research Project**
 - Designing the explainability and decision-support module of a multi-source yield forecasting system for Sri Lanka, combining remote sensing imagery with climate and field data in **Python**, **Pandas**, and **Scikit-learn** to give growers earlier, more accurate yield estimates.
 - Applying **SHAP** interpretability to expose the environmental and climatic drivers behind each prediction, turning a black-box model into transparent, evidence-backed guidance that stakeholders can trust.
 - Building farmer-friendly visualisations and a **RAG**-based advisory assistant that converts predictions into actionable, multilingual recommendations, closing the gap between model output and on-farm decisions.

TECHNICAL COMPETENCIES

- **Programming Languages:** Python, Java, C, C#, JavaScript, TypeScript
- **Web Development:** React, Next.js, Node.js, Express.js, FastAPI, REST APIs, HTML, CSS, Material UI, Tailwind CSS
- **AI & Intelligent Systems:** Generative AI, Agentic AI, AI Agents, Prompt Engineering, LLM Integration, AI-Assisted Development
- **Databases & Cloud:** MongoDB, MySQL, SQL, AWS
- **DevOps & CI/CD:** Docker, CI/CD Pipelines, GitHub Actions, Cloud Deployment
- **Core CS & Software Practices:** OOP, DSA, SDLC, RESTful API Design, Testing, Debugging
- **Tools & Platforms:** Git, GitHub, Postman (API Testing), Jupyter Notebook

CERTIFICATIONS & ACHIEVEMENTS

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| • React Basics - Meta (Coursera) | • Data Analytics - NoviTech R&D Pvt Ltd (Issued Jul 2024) |
| • Code Rush 2023 - INTECS, University of Moratuwa | • Idealize'24 - Participant |
| • Artificial Intelligence - NoviTech R&D Pvt Ltd (Issued Jul 2024) | • HackMoral 6.0 - Participant |

PROFESSIONAL REFERENCES

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